

Ranking Reversal Theory Under Candidate-Set Shift in GRN Benchmarking

Abstract

Gene regulatory network (GRN) benchmarks are routinely used to justify scientific claims about method quality, yet the stability of method *ranking* under plausible evaluation-protocol choices is rarely audited. We develop a framework for analyzing pairwise ranking reversals under protocol shift, supply a minimum-shift necessary condition for reversal, and quantify reversal behavior on existing single-cell GRN benchmark outputs across four protocol axes: candidate-set restriction, tissue, reference network, and gene-symbol mapping policy. Using cluster-bootstrap confidence intervals over method pairs — which correctly account for the dependence Wilson intervals miss — we find candidate-set shifts produce pairwise reversals in 16% of method pairs (cluster-bootstrap 95% CI 7-27%), tissue shifts in 19% (CI 9-31%), and reference-network shifts in 32% (CI 20-43%); mapping-policy shifts produce 0/165. The decomposition $\Delta = b \cdot g$ (base rate $\times$ discrimination) is structural and forces certain sign-attribution claims under positive base rates; we therefore report the genuinely empirical magnitude content: in every observed candidate-shift reversal the calibration-term magnitude exceeds the base-rate-term magnitude (22/22), versus only 56/113 in non-reversal rows (Mann-Whitney $p = 3.6 \times 10^{-7}$), with mean ratio 1.54 (median 1.02). A margin-magnitude sweep shows that no reversals occur above the 75th percentile of pair-margin magnitudes for candidate shifts, indicating the phenomenon is concentrated in near-tie pairs. AUROC produces 0/40 tissue-shift reversals where AUPR produces 4/45, suggesting reversal risk depends materially on metric choice. The practical recommendation: benchmark rank should not be interpreted as method-intrinsic evidence until protocol-axis stability and margin-magnitude regimes are reported.

1. Introduction

Mechanistic interpretability and benchmarking for biological models increasingly use leaderboard rank in GRN inference to justify claims about biological plausibility and downstream utility [1, 2]. The evaluation pipeline, however, involves several protocol choices — which candidate edges are scored, which reference network is used, how gene identifiers are mapped, and which tissue context is evaluated — that are rarely reported or controlled [3, 6].

If ranking is unstable under plausible protocol variation, downstream biological decisions can flip: which regulators are prioritized for experimental validation, which mechanistic narrative is emphasized, and which model is treated as scientifically credible. The field therefore needs explicit ranking-stability theory and diagnostics, not larger metric tables.

We make three contributions. (1) We define the pairwise reversal indicator under protocol shift and supply a minimum-shift necessary condition giving a closed-form “safety margin” per pair (Section 3). (2) We separate the algebraic content of the standard $\Delta = b \cdot g$ decomposition (which forces certain sign-attribution claims when base rates are positive) from its empirical content (the magnitudes of the two terms in reversal vs non-reversal rows), and report only the latter as findings (Section 5.2). (3) We quantify reversal behavior across four protocol axes with cluster-bootstrap confidence intervals over method pairs, three alternative null models, a per-axis margin-magnitude sweep, BH-FDR-adjusted per-cell rates, and a metric-robustness check (Sections 5.1–5.8). We treat the instability-region heuristic as a screening problem with an honest negative finding: neither a quantile heuristic nor a learned logistic-regression classifier exceeds baseline prevalence on this dataset (Section 5.7).

2. Related Work

GRN benchmarking and evaluation bias. Systematic GRN benchmarking has been addressed by [3, 4, 5], typically on fixed protocols. [6] showed that protocol choices such as symbol mapping and candidate-set restriction can shift AUPR by orders of magnitude, but focused on bias quantification rather than ranking stability. Our analysis is downstream: given those documented shifts, when does the leaderboard reorder?

Benchmark sensitivity in machine learning. The sensitivity of benchmark conclusions to evaluation choices is widely studied: BLEU rank instability under tokenization [Post 2018], benchmark-lottery effects [Dehghani et al. 2021], and rank stability under bootstrap [Demsar 2006]. We treat ranking reversal as a finite-margin event and supply an explicit minimum-shift bound, which the ML literature on rank stability has largely framed as average-case rather than worst-case.

Rank stability in statistics. Rank stability under sample perturbation has been studied via Kendall’s τ , top-k stability bounds, and Critchlow’s rank metrics. Our cluster-bootstrap CI over method pairs is the rank-stability analogue of cluster-robust inference on pairwise comparisons.

3. Problem Setup and Theory

3.1 Notation and reversal definition

Let $M_m(S, \pi, R)$ denote a scalar evaluation metric for method m under candidate set S , mapping policy π , and reference R . For methods A and B , define the **margin**:

$$\Delta = M_A - M_B.$$

For two protocol settings $(1, 2)$ with margins Δ_1, Δ_2 and shift $\delta\Delta = \Delta_2 - \Delta_1$, a **pairwise ranking reversal** is defined by $\Delta_1 \cdot \Delta_2 < 0$. Equivalently, $\text{sign}(\Delta_1) \cdot \delta\Delta < -|\Delta_1|$: the shift must oppose the initial ordering and exceed the initial margin. This is a definitional restatement, not a theorem; we include it because it makes the **safety margin** $|\Delta_1|$ explicit — a single pair cannot reverse under any shift family $\delta\Delta$ whose magnitude bound B satisfies $B \leq |\Delta_1|$.

3.2 Decomposition identity (definition)

For a fixed mapping policy and reference, the metric admits the trivial product form $\Delta(S) = b(S) \cdot g(S)$ with $g(S) := \Delta(S)/b(S)$ (assuming $b > 0$), where $b(S)$ is the candidate-set positive base rate and $g(S)$ is the base-rate-normalized discrimination gap. The finite product rule then gives an **exact** decomposition of $\delta\Delta$:

$$\Delta_2 - \Delta_1 = (b_2 - b_1) \cdot g_1 + b_2 \cdot (g_2 - g_1).$$

- *base-rate term*: $(b_2 - b_1) \cdot g_1$
- *calibration/discrimination term*: $b_2 \cdot (g_2 - g_1)$

This is an algebraic identity, not a theorem. We caution explicitly that **certain sign-attribution claims from this decomposition are structurally pinned, not empirical**, whenever both base rates are positive (as in every dataset slice we examine):

- The “only-b-changes” counterfactual $\Delta_1 \cdot (b_2 g_1) < 0$ simplifies to $(b_2/b_1) \Delta_1^2 < 0$, which is **always false** for $b_1, b_2 > 0$ and $\Delta_1 \neq 0$. So no observed reversal can ever be attributed to base-rate scaling alone — but this is a property of the decomposition under positive base rates, not a property of the data.
- The “only-g-changes” counterfactual $\Delta_1 \cdot (b_1 g_2) < 0$ simplifies to $(b_1/b_2) \Delta_1 \Delta_2 < 0$, which is **always true** on reversal rows (where $\Delta_1 \Delta_2 < 0$ by definition). So “discrimination change alone would have produced the reversal” holds 100% by construction.

The genuinely empirical content of the decomposition is the *magnitude* relationship between the two terms in reversal vs non-reversal

rows (Section 5.2): although the directions are pinned, the magnitudes are not. The minimum-shift bound (Proposition 1 below) gives the necessary inequality that any reversal must satisfy, and the empirical question is whether the calibration-term magnitude systematically exceeds the base-rate-term magnitude in reversal rows.

3.3 Minimum-shift necessary condition (Proposition 1)

Proposition 1. *For a shift family with bounded magnitude $|\delta\Delta| \leq B$, no pair with safety margin $|\Delta_1| > B$ can reverse. Furthermore, given the decomposition above, a necessary condition for reversal of a pair with margin $|\Delta_1|$ is*

$$|b_2 - b_1| \cdot |g_1| + b_2 \cdot |g_2 - g_1| \geq |\Delta_1|.$$

In particular, if g is stable under the shift ($|g_2 - g_1| \leq \varepsilon_g$), then the base-rate change required to reverse the pair satisfies $|b_2 - b_1| \geq (|\Delta_1| - b_2 \varepsilon_g) / |g_1|$. If b is stable ($|b_2 - b_1| \leq \varepsilon_b$), then the discrimination change required satisfies $|g_2 - g_1| \geq (|\Delta_1| - \varepsilon_b |g_1|) / b_2$.

Proof. The first claim is immediate from $|\Delta_2 - \Delta_1| \leq B$ and the reversal definition. The second follows from the decomposition and the triangle inequality applied to $\delta\Delta$. The corollaries follow by isolating each term.

Proposition 1 is the closed-form “is my pair safe?” calculator. Combined with the empirical distributions of $|\Delta_1|$ and $|g_2 - g_1|$ it yields per-pair safety bounds; the margin-threshold sweep in Section 5.6 visualizes the resulting safety regime by axis. Note that the corollaries are informative only when the right-hand sides are positive (e.g., the discrimination-stability corollary $|b_2 - b_1| \geq (|\Delta_1| - b_2 \varepsilon_g) / |g_1|$ is informative only when $b_2 \varepsilon_g < |\Delta_1|$; otherwise no constraint on $|b_2 - b_1|$ is implied).

3.4 Reference-shift remark

For the immune reference-shift slice, base rates differ across references by absolute amount $\approx 4.7 \times 10^{-4}$ (from $\approx 2 \times 10^{-6}$ for `beeline_gsd` to $\approx 4.8 \times 10^{-4}$ for `dorothea_trrust_union_immune`). Although the absolute range is non-trivial, the $b \cdot g$ decomposition is numerically unbalanced for reference shifts: empirically the mean magnitude ratio $|\text{calibration term}|/|\text{base-rate term}|$ is $\sim 10^4$ in both reversal and non-reversal rows (see `reference_shift_decomposition_summary.csv` from the prior pipeline).

With one term dominating by four orders of magnitude regardless of reversal status, the decomposition offers no discrimination between mechanisms here. We do not apply the $b \cdot g$ decomposition to reference shifts; we report reversal counts only. A coverage-and-precision decomposition ($M = c \cdot q$ with c = reference-overlap fraction, q = within-overlap quality) is a natural alternative; we use it for mapping shifts (Section 5.5) where overlap varies materially.

4. Methods

4.1 Datasets and methods compared

Inputs come from the evaluation-bias workshop study [6]: tissue-stratified medians of AUPR, AUROC, and F1 across three Tabula Sapiens tissues (kidney, lung, immune) and three candidate-set definitions (all_pairs, tf_sources, tf_sources_targets). The methods compared are the six in Table A3 of that study: **genie3**, **grnboost2**, **pearson**, **spearman**, **scgpt_attention**, **random**. Reference-shift analysis uses the immune score_eval_grn_baselines_immune slice with three references (hpn_dream, beeline_gsd, dorothea_trrust_union_immune) and nine methods. Mapping-policy analysis uses four policies (legacy_symbols, full_genes, crosswalk, omnipath_ref) on the probe-prior subset.

All inputs are summary medians; replicate-level raw predictions are not available in this slice. Uncertainty therefore reflects between-pair structural variability, not within-method replicate noise (see §8).

4.2 Analyses

For each of the four protocol axes we compute:

1. **Pairwise reversal indicator** $1\{\Delta_1 \cdot \Delta_2 < 0\}$ per (method-pair, shift-cell).
2. **Cluster-bootstrap 95% CIs** with the method pair as the cluster, using 2,000 resamples of clusters; this addresses dependence Wilson intervals miss [reviewer concern C5].
3. **Magnitude decomposition** (candidate shifts only): per reversal row, compute the calibration-vs-base-rate magnitude ratio and the Mann-Whitney comparison against non-reversal rows. We do not report sign-attribution counts because they are algebraically pinned under positive base rates (§3.2).
4. **Per-cell BH-FDR adjustment** on highlighted shift cells (two-sided test against rate = 0.5).
5. **Margin-magnitude threshold sweep**: reversal rate as a function of τ , the minimum of $|\Delta_1|$, $|\Delta_2|$, with cluster-bootstrap bands.

6. **Three null models** (candidate shift only): (a) joint method-relabel in c_2 , (b) score-noise bootstrap with $\sigma = 10\%$ of within-cell median absolute margin, (c) rank permutation within (tissue $\times$ candidate) cell. (a) and (c) are the natural “is the rank signal real?” nulls; (b) targets the question “would these reversals have happened anyway under plausible AUPR noise?”.
7. **Instability screening**: leave-one-tissue-out classifier evaluation, comparing the quantile heuristic (predicted unstable iff $|\Delta_1| \leq \text{quantile_q of } |\delta\Delta|$) against a logistic regression learned on $(|\Delta_1|, b_1, b_2 - b_1, |g_1|)$ features. Report PR-AUC, not only operating-point precision/recall.
8. **Metric robustness**: repeat the tissue-shift analysis using AUPR, AUROC, and F1 (Table A1).

All numerical decomposition closure errors are $< 1e-9$ (verified in code; tolerance threshold checked at runtime). All seeds are fixed (project seed 42; permutation seed 42; bootstrap seed 42).

5. Results

A summary table of all four axes is given as Table 1.

Table 1. Headline reversal rates by axis (cluster-bootstrap 95% CIs)

Axis	n pairs	k reversals	Rate	Cluster CI	Metric	Dataset slice
Candidate-set shift (cand-conditional)	135	22	0.16	[0.07, 0.27]	AUPR	A2/A3 (3 tissues $\times$ 3 cand $\times$ 6 methods)
Tissue shift (cand-conditional)	135	26	0.19	[0.09, 0.31]	AUPR	A3 (3 tissues $\times$ 3 cand $\times$ 6 methods)

Axis	n pairs	k reversals	Rate	Cluster CI	Metric	Dataset slice
Tissue shift (dedup, any candidate)	45	17	0.38	[0.18, 0.58]	AUPR	A3 (one row per method-pair $\times$ tissue-pair)
Tissue shift (dedup, all candidates)	45	1	0.02	[0.00, 0.07]	AUPR	A3
Reference shift	106	34	0.32	[0.20, 0.43]	AUPR	immune GRN base-lines, 9 methods, 3 refs
Mapping-policy shift	165	0	0.00	[0.00, 0.00]	F1	probe priors, 4 policies

Numerical precision is rounded to two decimals consistent with cluster-CI half-widths. The tissue-shift axis is reported with three rates because they answer different questions: the candidate-conditional 0.19 counts each method-pair $\times$ tissue-pair $\times$ candidate triple separately (and is therefore inflated by dependent observations); the dedup-any 0.38 counts a method-pair $\times$ tissue-pair as a reversal if it reverses under any candidate set; the dedup-all 0.02 counts a method-pair $\times$ tissue-pair only if it reverses under all candidate sets simultaneously. The honest single-number summary is the dedup-any rate, 0.38; we report all three for transparency.

Candidate-shift reversal rate by tissue $\times$ transition

Figure 1: Candidate-shift reversal rate by tissue $\times$ transition

5.1 Candidate-set shifts produce nontrivial reversal risk concentrated in near-tie pairs

Evidence. Candidate-set shifts produce 22/135 reversals (rate 0.16, cluster-bootstrap 95% CI [0.07, 0.27]). Per-cell BH-FDR-adjusted rates (candidate_shift_cells_fdr.csv) show that **none** of the high-reversal cells highlighted in earlier versions of this analysis (e.g., immune all_pairs $\rightarrow$ tf_sources_targets at 6/15 = 0.40) survive $q < 0.05$ as significantly above 0.5; what survives is the opposite — cells significantly *more stable* than chance (kidney tf_sources $\rightarrow$ tf_sources_targets at 0/15, $q = 5 \times 10^{-4}$; lung all_pairs $\rightarrow$ tf_sources at 0/15, $q = 5 \times 10^{-4}$).

Inference. Rank is not a method-intrinsic invariant; it is protocol-conditional. The marginal 16% rate is driven by the immune tissue and small-margin pairs (Section 5.6).

Scientific implication. A single leaderboard rank should not be used to justify biological claims without candidate-shift stability reporting, but the per-cell tail rates in the previous version of this analysis were not robust to multiple testing.

5.2 Calibration magnitude exceeds base-rate magnitude in every observed reversal (empirical)

Structural note (what is *not* a finding). As shown in Section 3.2, under positive base rates the decomposition $\Delta = b \cdot g$ algebraically forces (i) the calibration term to have sign opposite to Δ_1 in every reversal row, and (ii) the base-rate term to have sign aligned with Δ_1 . These are properties of the decomposition, not of the data. We therefore do **not** report sign-attribution counts (0/22, 22/22, etc.) as findings; doing so was the central flaw of earlier versions of this analysis.

Evidence (genuinely empirical). What is *not* structurally pinned is the relative *magnitude* of the two terms. In our reversal rows we observe:

- |calibration term| > |base-rate term| in **22/22** reversal rows (rate 1.00), versus **56/113** in non-reversal rows (rate 0.50).
- Mean magnitude ratio |cal|/|base-rate| in reversal rows: **1.54**; median: **1.02**.
- Mean ratio in non-reversal rows: **0.83**; median: **1.00**.

Calibration vs base-rate magnitude ratio in reversal vs non-reversal rows

Figure 2: Calibration vs base-rate magnitude ratio in reversal vs non-reversal rows

- Mann-Whitney U test (one-sided: reversal-row ratios stochastically larger than non-reversal): $U = 2075$, $p = 3.6 \times 10^{-7}$.
- The necessary condition from Proposition 1 — specialized to our positive-base-rate setting as $|T_{cal}| - |T_{br}| > |\Delta_1|$ — is satisfied in 22/22 reversal rows (by construction, since this is what *defines* a reversal under the decomposition).

Inference. The empirical content of the decomposition is captured by the magnitude distribution, not by sign tallies. Reversal rows exhibit a calibration-magnitude advantage that is (a) universal in our data (22/22 with $|cal| > |base-rate|$) and (b) statistically much larger than in non-reversal rows. Because the magnitude inequality is *not* algebraically pinned by reversal status (a hypothetical reversal could in principle have $|base-rate| > |cal|$ under negative base rates or different shift directions), this is a meaningful data finding rather than a re-statement of the decomposition.

Hypothesis (biological). Methods differ in their within-candidate-space discrimination of biologically plausible vs implausible edges; the magnitude evidence is consistent with discrimination shifts being the dominant *driver* of observed reversals in this slice, though sign causality is structural, not empirical.

Scientific implication. Benchmark interpretation should report the calibration-vs-base-rate magnitude ratio when discussing rank stability, not the sign-attribution counts that the decomposition algebraically pins.

5.3 Tissue shifts: deduplicated rate is much larger than the candidate-conditional rate suggests

Evidence. Treating each (tissue_from, tissue_to, method_a, method_b) triple as one observation (dedup), 17 of 45 method-pair $\times$ tissue-pair triples reverse under at least one candidate-set choice (rate 0.38, cluster CI [0.18, 0.58]). Only 1/45 reverse under *all* candidate-set choices (rate 0.02). The candidate-conditional rate of 0.19 reported in the previous version of this analysis lies between these two and inflates the underlying signal by counting each pair three times.

Inference. Cross-tissue rank transportability is real but the magni-

Tissue-shift rate by candidate set (kept from prior pipeline)

Figure 3: Tissue-shift rate by candidate set (kept from prior pipeline)

Reference-shift reversal rate (kept from prior pipeline)

Figure 4: Reference-shift reversal rate (kept from prior pipeline)

tude depends sensitively on the reporting convention. The honest single-number summary is dedup-any = 0.38.

Hypothesis (biological). Candidate constraints amplify tissue-specific regulon mismatch by reducing background edge averaging; the 0.38 vs 0.02 gap shows most tissue-shift reversals are candidate-conditional, not universal.

Scientific implication. Cross-tissue mechanistic claims should report rank transportability under each candidate-set definition separately.

5.4 Reference shifts produce robust reversals across margin thresholds

Evidence. Reference-network shifts in the immune slice produce 34/106 reversals (rate 0.32, cluster CI [0.20, 0.43]). Per-cell BH-FDR: *dorothea_trrust_union_immune* → *hpn_dream* at rate 0.19 is significantly *below* 0.5 ($q = 7e-4$); the previously-highlighted *beeline_gsd* → *dorothea_trrust_union_immune* cell at 0.43 does **not** survive BH-FDR ($q = 0.40$) — its rate is not statistically distinguishable from 0.5 at this sample size. Importantly, the margin-magnitude sweep (Figure 2) shows reference-shift reversals **persist** across all margin thresholds (rate stays in [0.22, 0.38] up to the 75th percentile of margins), while candidate-shift reversals vanish at high margins (0/34 at the 75th-percentile candidate threshold).

Inference. Reference choice is a robust instability source: unlike candidate-shift instability, it is not concentrated in near-tie pairs.

Scientific implication. Single-reference “best method” claims are likely overconfident; multi-reference sensitivity should be standard.

5.5 Mapping-policy shifts changed coverage without reversing pairwise order

Evidence. Mapping-policy shifts show 0/165 pairwise reversals (Wilson upper 95% bound 2.3%, cluster bootstrap [0.00, 0.00]). Legacy-symbols → full-genes/crosswalk increases mean coverage by +0.86 (absolute) with near-zero mean F1 shift ($-4e-5$).

Reversal rate vs margin threshold

Figure 5: Reversal rate vs margin threshold

Inference. In this subset, mapping behaves as an approximately order-preserving transform despite large overlap gain.

Hypothesis (biological). The probe-prior subset may be less sensitive to alias ambiguity than broader evaluations with heterogeneous identifier conventions; this should not be generalized.

Scientific implication. Coverage changes still require reporting because they alter interpretability and comparability even when ordering is unchanged.

5.6 Margin magnitude controls reversal risk on the candidate axis but not the reference axis

Evidence. Figure 2 reports reversal rate as a function of the margin-magnitude floor $\tau = \min(|\Delta_1|, |\Delta_2|)$. For candidate shifts, rate drops from 0.16 ($\tau = 0$, all 135 pairs) to 0.00 ($\tau = 75\text{th-percentile margin}$, 34 pairs kept). For reference shifts, rate remains in $[0.22, 0.38]$ up to the 75th-percentile margin. For tissue shifts, rate decays gradually from 0.19 to 0.09. For mapping shifts the rate is 0 throughout.

Inference. Candidate-shift “instability” is overwhelmingly a phenomenon of near-tie method pairs; on pairs with margins above the median, candidate-shift reversal is exceedingly rare. Reference-shift instability is genuinely method-pair-structural and not explained by near-ties.

Scientific implication. Reporting reversal rates without a margin-magnitude regime is ambiguous; future stability reports should always include a margin sweep.

5.7 Reversal behavior is structured across three null models, but instability screening fails to beat baseline prevalence

Evidence. Three null models for the candidate-shift reversal rate are compared in Figure 3:

Observed rate vs three nulls

Figure 6: Observed rate vs three nulls

Instability screening curves

Figure 7: Instability screening curves

Null	Null mean (95% interval)	Observed
Joint method-relabel in c_2	0.50 [0.37, 0.61]	0.16
Score-noise bootstrap ($\sigma = 10\%$ of median margin)	0.29 [0.21, 0.37]	0.16
Rank permutation within (tissue $\times$ candidate) cell	0.50 [0.37, 0.61]	0.16

The score-noise null is a perturbation analysis rather than a true null: it adds Gaussian noise of magnitude $\sigma = 10\% \times$ (within-cell median pairwise margin) to each observed score and recomputes the reversal rate. The reported value 0.29 is the *noise-inflated* rate; the observed 0.16 is the noise-free rate. Adding noise increases reversals (more pairs are perturbed across the sign-flip boundary), so $0.29 > 0.16$ is mechanically expected; the substantive reading is that observed reversals are *robust* to $\sim 10\%$ -of-margin-scale score noise, i.e., the underlying ordering structure is tighter than that noise scale and the 16% reversal rate is not a noise artifact. The two relabel nulls confirm the rank signal is far from random.

For the instability-screening problem, the leave-one-tissue-out quantile heuristic peaks at precision 0.24, recall 0.64, F1 0.35 at threshold 0.25, and produces a PR-AUC of 0.18 — only marginally above the baseline reversal prevalence 0.16. A learned logistic-regression classifier on $(|\Delta_1|, b_1, b_2 - b_1, |g_1|)$ features yields PR-AUC 0.13, *below* baseline. This is an honest negative result: neither classifier provides actionable lift on this dataset slice.

Inference. Rank signal is real (well-separated from three independent nulls) but per-pair instability is not predictable from these features on this dataset; either richer features or replicate-level training signal would be needed.

Metric-robustness summary

Figure 8: Metric-robustness summary

5.8 Reversal risk depends materially on metric choice (AUROC is most stable)

Evidence. Repeating the tissue-shift analysis on Table A1 (no candidate stratification) with three metrics:

Metric	n	k	Rate	Wilson 95% CI
AUPR	45	4	0.09	[0.04, 0.21]
F1	45	4	0.09	[0.04, 0.21]
AUROC	40	0	0.00	[0.00, 0.09]

Inference. AUROC produces no observed tissue-shift reversals in this dataset, while AUPR and F1 both produce 4 reversals (the same four pairs in each case). The choice of metric matters for the rank-stability conclusion.

Hypothesis. Because AUROC is invariant to monotone transformations of the score and AUPR is not, AUROC’s stability reflects the underlying score-rank similarity across tissues; AUPR’s instability reflects sensitivity to where the operating-rank-density lies relative to the positive base rate.

Scientific implication. Stability claims should be reported per metric; “method X is better on tissue Y” is metric-conditional and rank-conditional.

6. Biological Interpretation

The core biological lesson is that benchmarking protocol defines the biological question being asked. Candidate-space restrictions emphasize particular regulator-target submanifolds, reference changes reweight biological evidence classes, and tissue context changes the active regulatory program. Observed method rank mixes algorithmic quality with biological framing choices. We constrain our claims to protocol-conditional benchmarking behavior, not causal biological discovery; the hypothesis blocks in §5 are *conjectures* for follow-up, not tested findings of this paper.

7. Discussion

Three patterns persist across our analyses. (i) Among reversal rows, the calibration-term magnitude exceeds the base-rate-term magnitude in 22/22 cases (vs 56/113 in non-reversal rows; Mann-Whitney $p = 3.6 \times 10^{-7}$), establishing on a magnitude basis (the only basis on which the $b \cdot g$ decomposition carries empirical content under positive base rates) that discrimination dominance accompanies the reversal phenomenon in this slice (§5.2). (ii) Candidate-shift instability is concentrated in near-tie pairs (§5.6); reference-shift instability is not. (iii) Metric choice materially affects the rank-stability conclusion: AUROC is fully stable across tissues in this slice while AUPR is not (§5.8).

We are deliberate about what is *not* a finding here. The “discrimination term opposes initial margin in 100% of reversals” sign-tally and the analogous “0/22 base-rate-only counterfactual” are algebraic consequences of the decomposition $\Delta = b \cdot g$ under positive base rates and a fixed shift direction; they re-state the reversal definition rather than evidence about the mechanism (§3.2). Reporting them as findings, as earlier versions of this analysis did, conflates structure with data.

Our instability-screening results are honestly negative: neither a quantile heuristic nor a learned classifier outperforms baseline prevalence. We interpret this not as failure of the framework but as a sign that protocol-conditional reversal is, at this data size, a population-level phenomenon rather than a per-pair predictable one. Stronger screening would likely require replicate-level uncertainty estimates and a larger method $\times$ tissue $\times$ reference grid.

8. Limitations

1. Inputs are summary medians; raw per-replicate AUPR is not exposed by the source artifacts, so within-method uncertainty is not propagated into our CIs. Cluster bootstrap addresses dependence across pairs but not within-cell measurement noise.
2. Reference-shift analysis is centered on the immune baseline slice (one tissue, three references); the reference-shift conclusions should not be extrapolated to other tissues without re-running.
3. Mapping-policy results are limited to the probe-prior subset; the 0/165 finding does not imply mapping is irrelevant in general, only in this slice.
4. The minimum-shift bound (Prop. 1) is a necessary condition; we do not prove sufficiency.
5. Biological hypotheses in §5 are conjectures, not tested findings.

9. Data and Code Availability

All code, source data, regenerated tables, and figures used in this paper are released at the (author-anonymized) repository:

<https://github.com/Biodyn-AI/grn-ranking-reversal-theory>

Repository layout: data/ (Table A1/A2/A3 summary medians + the immune GRN baseline and four mapping-policy CSVs), scripts/analysis.py (self-contained pipeline), results/tables/ and results/figures/ (regenerated outputs), paper/ (manuscript source and compiled PDF). Reproducing the analysis end-to-end requires Python 3.9+ and the packages listed in requirements.txt:

```
pip install -r requirements.txt
python scripts/analysis.py
```

All random seeds are fixed (project seed 42; permutation seed 42; bootstrap seed 42). Numerical decomposition closure is verified at runtime to within $1e-9$.

10. Conclusion

Ranking reversal is a first-order reliability issue for GRN benchmarking. Cluster-bootstrap CIs that respect pair-level dependence, margin-magnitude sweeps, per-axis BH-FDR-adjusted cell rates, and a magnitude-based reading of the $b \cdot g$ decomposition (rather than the structurally pinned sign tallies) together replace the standard “rate $\pm$ Wilson interval” report with a more honest characterization of what is robust and what is not. The practical recommendation is direct: treat method rank as biologically interpretable evidence only after cross-axis stability is demonstrated, the reversal-prone margin regime is characterized, and the choice of metric is reported.

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